Professor Arun Majumdar

- **B.Tech in Mechanical Engineering from the Indian Institute of Technology in 1985**
- **PhD in Mechanical Engineering from the University of California, Berkeley in 1989.**
- **Faculty of Arizona State University (1989-92)**
- **Faculty in University of California, Santa Barbara (1992-96)**
- **Faculty in University of California, Berkeley from January 1, 1997.**

热力学与统计物理

White House nominates Berkeley Lab's Majumdar to head key DOE agency

SEPTEMBER 18, 2009

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:: NEWS RELEASE

Contact: Jeff Miller, jwmiller@lbl.gov, (510) 508-1090

Berkeley, CA — The White House announced its intention to nominate Lawrence Berkeley National Laboratory and University of California, Berkeley engineer and scientist Arun Majumdar to become the first director of the U.S. Department of Energy's Advanced Research Projects Agency-Energy (ARPA-E).

Berkeley Lab's Arun Majumdar has been nominated to head the U.S. Department of Energy's Advanced Research Projects Agency-Energy.



ARPA-E's goals are to create technologies that have the potential to reduce the nation's reliance on foreign energy supplies, reduce energy-related greenhouse gas emissions, and improve energy efficiency. It focuses on high-risk, high-payoff technologies that promise to transform the energy landscape.

Majumdar's nomination must undergo Senate confirmation.

A mechanical engineer and materials scientist, Majumdar's wide-ranging expertise spans energy efficiency technology research and implementation in the U.S. and abroad, as well as materials science innovations such as using nanotechnology to harness the energy lost as heat during the production of electricity.

1. Duan and A. Majumdar, “Anomalous Ion Transport in 2-nm Hydrophilic Nanochannels”, Nature Nanotechnology, Vol. 5, pp 848-852, (2010)

2. A. I. Hochbaum*, R. Chen*, R. D. Delgado, W. Liang, E. C. Garnett, M. Najarian, A. Majumdar and P. Yang, “Enhanced Thermoelectric Performance of Rough Silicon Nanowires,” Nature, Vol.451, pp. 163-167, (2008)

3. P. Reddy, S. Y. Jang, R. A. Segalman, A. Majumdar. "Thermoelectricity in molecular junctions," Science, Vol. 315, pp. 1568-1571, (2007)

4. C. W. Chang, D. Okawa, A. Majumdar, A. Zettl. “Solid-State Thermal Rectifier,” Science, Vol. 314, pp. 1121-1124 (2006).

5. A. Majumdar, “Thermoelectricity in Semiconductor Nanostructures,” Science, Vol. 303, pp. 777-778 (2004).

6. G. Wu, R. Datar, K. M. Hansen, T. Thundat, R. J. Cote, A. Majumdar, “Bioassay of Prostate Specific Antigen (PSA) Using Microcantilevers,” Nature *Biotechnology*, Vol. 19, pp. 856-860 (2001).

7. A. Majumdar, “Scanning Thermal Microscopy,” Annual Rev. Materials Science, Vol. 29, pp. 505-585 (1999).



Gang Chen

- Carl Richard Soderberg Professor of Power Engineering, MIT
- Director, Pappalardo Micro and Nano Engineering Laboratories
- Director, DOE EFRC: Solid-State Solar-Thermal Energy Conversion Center
- B.S., 1984, M.S. 1987, Huazhong University of Science and Technology
- 1993 Ph.D. UC Berkeley
- 1996 Associate Professor, UCLA
- 1993 Assistant Professor, Duke University
- 1987 Lecturer, Huazhong University of Science and Technology, China.

MITnews

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Breaking the law, at the nanoscale

Bringing objects close together can boost radiation heat transfer, according to new study that shows breakdown in Planck's law

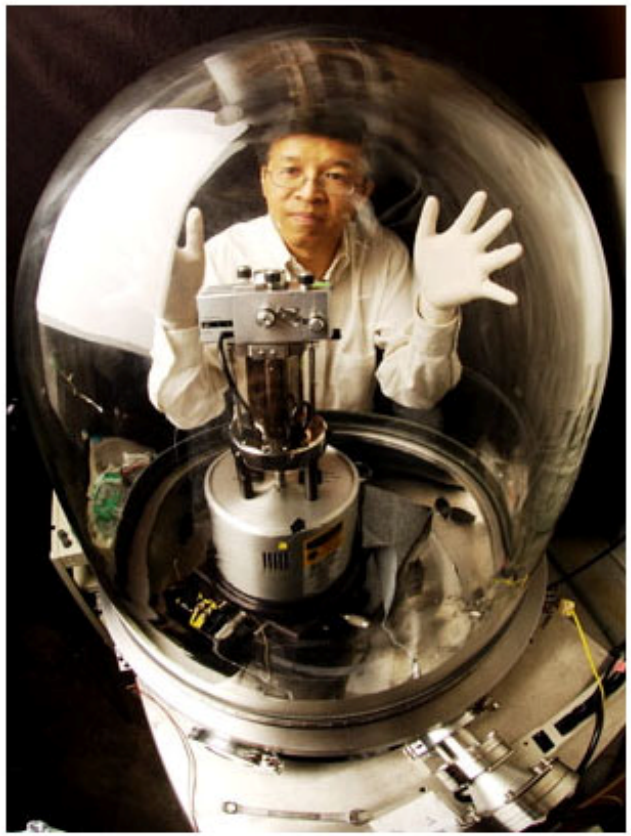
David L. Chandler, MIT News Office

July 29, 2009

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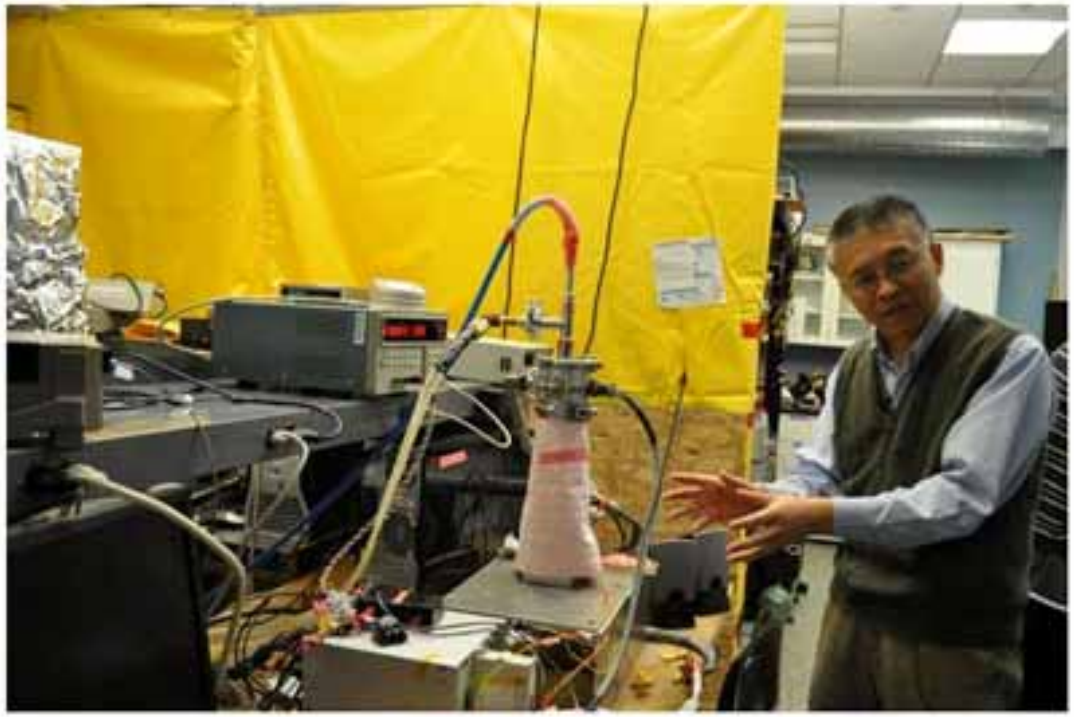
A well-established physical law describes the transfer of heat between two objects, but some physicists have long predicted that the law should break down when the objects are very close together. Scientists had never been able to confirm, or measure, this breakdown in practice. For the first time, however, MIT researchers have achieved this feat, and determined that the heat transfer can be 1,000 times greater than the law predicts.

The new findings could lead to significant new



Professor Gang Chen with the vacuum chamber used in this research.
Courtesy / Gang Chen

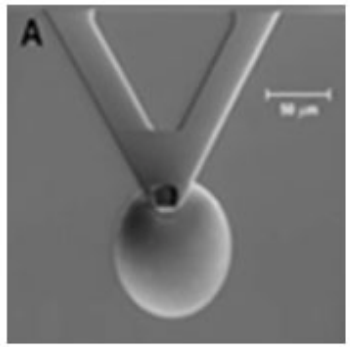
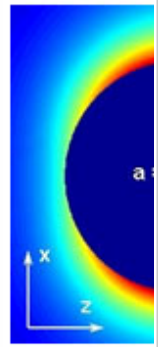
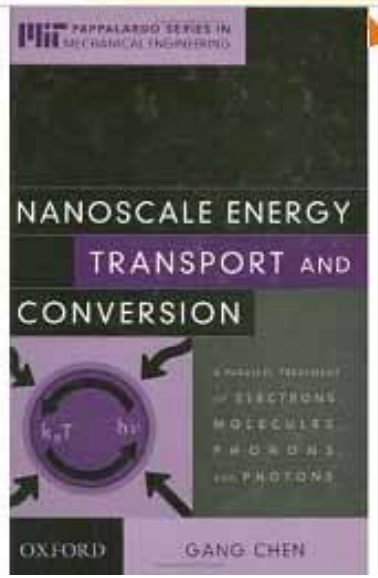
Pushing the boundaries of energy—theoretical, practical, and personal



Gang Chen—one of the pioneers on the front lines of energy science—is an energy phenomenon himself. The Carl Richard Soderberg Professor of Power Engineering is one of the world's leading researchers in the nanoengineering of energy transfer. He heads the new Solid State Solar Thermal Energy Conversion Center (S^3TEC) with a staff of 50+, serves on the editorial boards of five scientific journals, teaches undergraduate and graduate classes, advises 21 graduate students, 12 postdocs, and two research scientists, and delivers lectures and seminars from one end of the planet to the other. And in whatever time is left at the end of the day, Gang Chen is helping to redefine the laws of physics.

NANO :: HEAT :: ENERGY



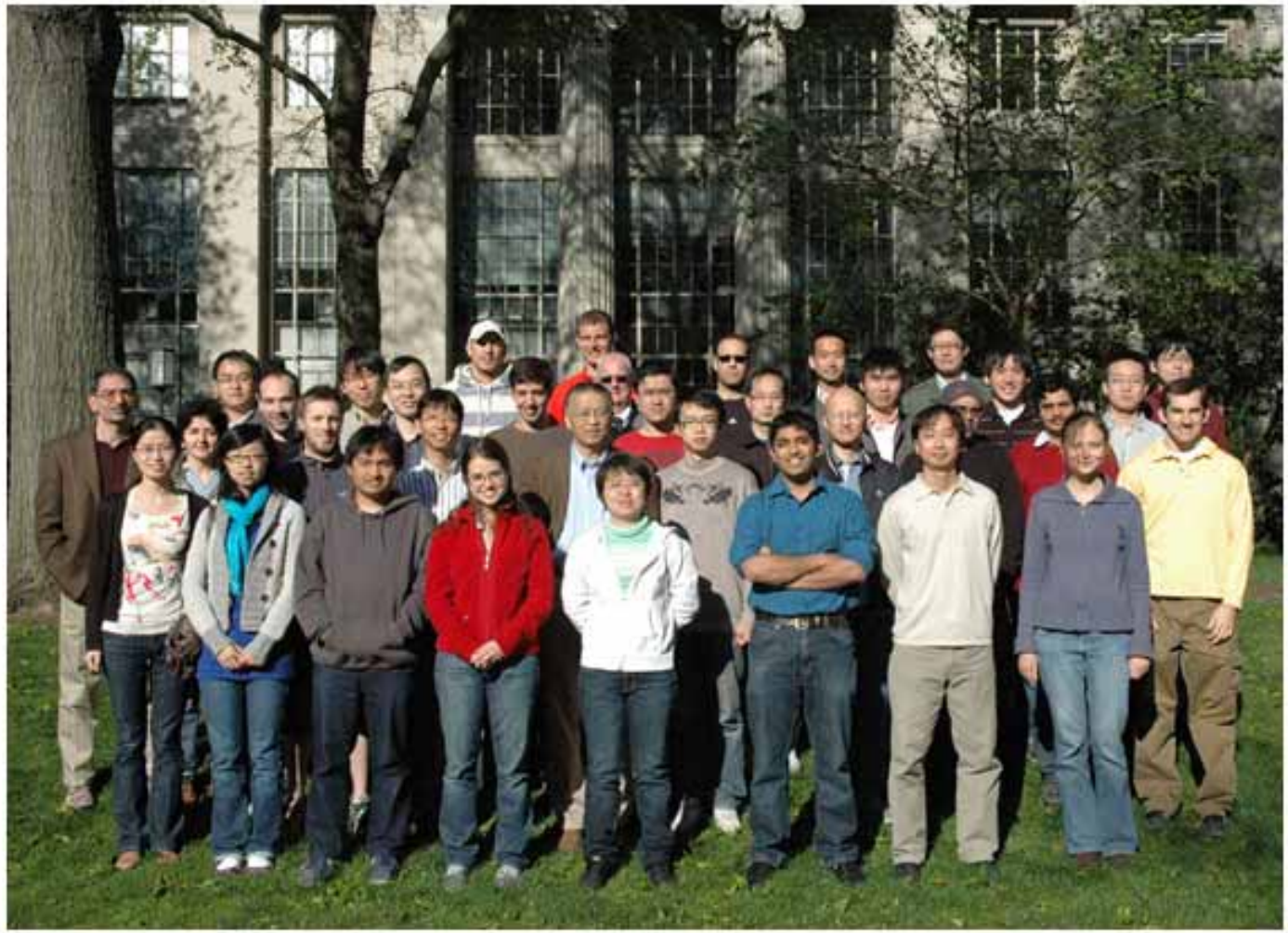
The Nanoengineering Group is part of the Mechanical Engineering Department at MIT. Our research is focused on nanoscale energy transport, conversion, and storage. There are fundamental differences between transport processes at the nanoscale and the macroscale due to quantum and classical size effects; for example, both classical diffusion laws and Planck's law for blackbody radiation break down in nanostructures. We explore these effects for improving energy conversion efficiency and storage density, and thermal energy transport. Examples include development of nanostructured thermoelectric materials for direct conversion between thermal and electric energy, use of nanotechnology to advance solar thermal and solar photovoltaic devices, fundamental investigation of phonon transport. Our work is interdisciplinary and combines mechanical engineering, electrical engineering, physics, and materials science.

NanoEngineering Group

MASSACHUSETTS INSTITUTE OF TECHNOLOGY

Members

- Faculty and Associates
- Students
- Visitors and Postdocs
- Alumni



The Nanoengineering Group, Spring 2010

Professor Gang Chen elected as APS Fellow

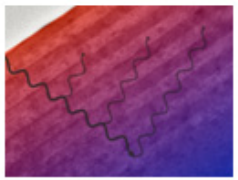
Professor Gang Chen was recently elected as APS Fellow in 2012. Official quote: "For pioneering contributions to the understanding of heat transfer at nanoscale and to the development of thermoelectric energy conversion technologies". Updated Nov. 29, 2012.

Directional Solvents Desalination Among Top 10 World Changing Ideas 2012, picked by Scientific American



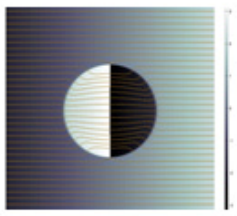
Directional solvents desalination, a new promising method of treating dirty water from fracking for natural gas and oil, was selected as one of Top 10 World Changing Ideas 2012, by Scientific American. Updated Nov. 21, 2012.

MIT News: Controlling heat flow through a nanostructure



Nanoengineering researchers found that heat moving in materials called superlattices behaves like waves. This finding could enable better thermoelectrics. This work was published on Science recently and you can check out the full news article here.

MIT News: "Invisibility" could be a key to better electronics



Researchers in NanoEngineering applied technology developed for visual "cloaking" to enable more efficient transfer of electrons. This work was published on Physics Review Letters and highlighted by Physics Synopsis and Nature Nanotechnology. It also gained some media coverage, such as in MIT news and PhysicsWorld, and was chosen as the cover story of MIT's website on Oct. 15, 2012.

Congratulations MIT NanoEngineering class of 2011

Congratulations to Andy Muto and Austin Minnich on receiving their Ph.D. degrees. Dr. Minnich is joining the mechanical engineering faculty at CalTech and Dr. Muto has taken an industry job with Corning. In addition to the graduates, congratulations are in order for the five post-docs leaving our group. Dr. Nitin Shukla is joining Fraunhofer Institute as a thermal engineer in their building technology group; Dr. Hsien-Ping Feng will be an assistant professor at Hong Kong University; Shuo Chen has been appointed to a senior research associate at Boston College; Dr. Celine Hin is joining the faculty at Virginia Tech; and Dr. Tengfei Luo will be a professor at the University of Notre Dame.

Solar Thermoelectric Generators Making News



Thermoelectric devices, which convert heat to electricity, can be used to convert solar power into electrical power. The devices comprise a surface which absorbs the sun's heat; a heat sink that stays at a fixed temperature; and a thermoelectric material between the absorber and the heat sink which converts this temperature difference into an electrical current. The recent NanoEngineering Group paper by Kraemer et. al, published in Nature Materials, has garnered a good deal of press. Read coverage from MIT News or The Economist.

MIT News: Nanofluid Thermal Conductivity

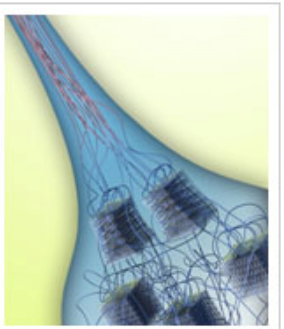


Researchers in Prof. Chen's lab have demonstrated the ability to control the thermal and electrical conductivities of fluids by creating networks of graphite in the solution. Read the full article here.

DOE establishes EFRC under Professor Gang Chen

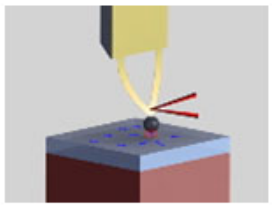
The U.S. Department of Energy has awarded $17.5M to create an Energy Frontiers Research Center (EFRC) at MIT under the direction of Professor Gang Chen. The Solid-State Solar Thermal Energy Conversion Center (S3TEC) will work to create solid-state materials to convert sunlight and heat into electricity.

MIT News: Insulators made into conductors



Sheng Shen, Asegun Henry, Jonathan Tong, Ruiting Zheng, and Professor Gang Chen have shown that polymers can be as good conductors as many metals. Their letter, published March 7 in Nature Nanotechnology, describes how drawing polymer fibers causes molecular chain alignment. This alignment leads to the anisotropically high thermal conductivity while maintaining low electrical conductivity. Materials with such properties could be used to solve many thermal management problems. Read the full article in MIT News or Nature.

MIT News: Breaking the law, at the nanoscale



Sheng Shen, Arvind Narayanaswamy, and Professor Gang Chen have proven it possible to break Planck's blackbody radiation law when the distance between objects is on the nanoscale. Careful measurements have shown the heat transfer between two bodies to exceed 1000 times what is predicted by Planck's law. Read the full article in MIT News or Nature.

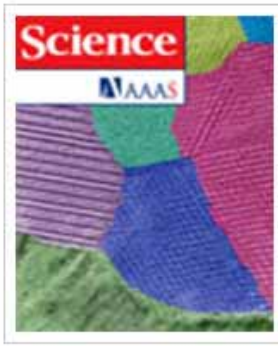
Nature Paper: Superplastic Carbon Nanotubes



Conditions have been discovered that allow extensive deformation of rigid single-walled nanotubes.

J. Y. Huang, S. Chen, Z. Q. Wang, K. Kempa, Y. M. Wang, S. H. Jo, G. Chen, M. S. Dresselhaus, Z. F. Ren, "Superplastic carbon nanotubes," *Nature* **439**, 281-281 (19 Jan 2006). Full article

Science Express Paper: High-Thermoelectric Performance of Nanostructured Bismuth Antimony Telluride Bulk Alloys



Here we show that a peak ZT of 1.4 at 100°C can be achieved in p-type nanocrystalline bismuth antimony telluride bulk alloy. These nanocrystalline bulk materials were made by hot-pressing nanopowders ball-milled from crystalline ingots under inert conditions. More importantly, ZT is about 1.2 at room temperature and 0.8 at 250C, which makes these materials useful for cooling and power generation. This discovery sets the stage for use of a new nanocomposite approach in developing high performance low-cost bulk thermoelectric materials. Full article

Ronggui Yang Wins the 2005 Goldsmid Award



From the International Thermoelectric Society:
"Ronggui Yang of the Massachusetts Institute of Technology met demanding standards to win the 2005 Goldsmid Award for Excellence in Research in Thermoelectrics by a Graduate Student. The Goldsmid Award, presented to Mr. Yang at the ICT2005 banquet, seeks to identify exceptional Ph.D. candidate graduate students who have made unusually notable contributions to thermoelectrics. Mr. Yang's publication record is impressive for anyone, much less for a student." Read more

1. M. N. Luckyanova, J. Garg, K. Esfarjani, A. Jandl, M. T. Bulsara, A. J. Schmidt, A. J. Minnich, S. Chen, M. S. Dresselhaus, Z. F. Ren, E. A. Fitzgerald and G. Chen, Coherent Phonon Heat Conduction in Superlattices, Science, 338, 936, 2012.

2. D. Kraemer, B. Poudel, H.P. Feng, J.C. Caylor, B. Yu, X. Yan, Y. Ma, X.W. Wang, D. Wang, A. Muto, K. McEnaney, M. Chiesa, Z.F. Ren, and G. Chen, High-performance flat-panel solar thermoelectric generators with high thermal concentration, Nature Materials, Vol. 10, pp. 532-538, 2011.

3. R.T. Zheng, J. Gao, J. Wang, and G. Chen, Reversible temperature regulation of electrical and thermal conductivity using liquid-solid phase transitions, Nature Communications, Vol. 2, 289, 2011.

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5. Bed Poudel, Qing Hao, Yi Ma, Yucheng Lan, Austin Minnich, Bo Yu, Xiao Yan, Dezhi Wang, Andrew Muto, Daryoosh Vashaee, Xiaoyuan Chen, Junming Liu, Mildred S. Dresselhaus, Gang Chen, and Zhifeng Ren, High Thermoelectric Performance of Nanostructured Bismuth Antimony Telluride Bulk Alloys, Science, Vol. 320, pp. 634-638, May 2, 2008.

6. J. Y. Huang, S. Chen, Z.Q. Wang, K. Kempa, Y. M. Wang, S. H. Jo, G. Chen, M.S. Dresselhaus, and Z. F. Ren, Superplastic single-walled carbon nanotubes, Nature, Vol. 439, p. 281, 2006.



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Associate Fellow of the American Institute of Aeronautics and Astronautics

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RESEARCH GROUP

Profile

Research focuses on the development of new technologies and the understanding of basic principles in the areas of sustainable energy, propulsion, and functional nano-materials. Current research projects comprise four primary areas:

- Microscale energy conversion via combustion and catalytic reactions
- Alternative fuels for internal combustion engines and propulsive systems

- 1981年安徽省桐城中学
- 1986年清华大学工程力学系，学士
- 1988年清华大学工程力学系，硕士
- 1988-90年清华大学工程力学系，博士研究生
- 1990-94年日本东北大学机械航空系， 博士（工学）
- 1994-95年日本东北大学流体科学研究所，助教
- 1995-98年日本东北大学流体科学研究所，讲师
- 1998-2000年日本东北大学机械航空系，副教授
- 2000-2001年清华大学长江特聘教授，工程热物理研究所所长
- 2001-2007年美国普林斯顿大学，助理教授
- 2007-2011年美国普林斯顿大学，副教授
- 2011-至今美国普林斯顿大学，教授，可持续能源专科主任



Members

- Principal Investigator
- Postdoc
- Graduate Students
- Visiting Students
- Undergraduate

Alumni

Principal Investigator



Deyu Li

deyu.li@vanderbilt.edu

Deyu Li is an associate professor in the department of Mechanical Engineering at Vanderbilt. He received his B.E. in Engineering Thermophysics from the University of Science and Technology of China (1992), M.E. in Mechanical Engineering from Tsinghua University, China (1997) and Ph.D. in Mechanical Engineering from UC Berkeley (2002), respectively. He worked as a post-doctoral research associate at UC Berkeley in 2003. He joined the faculty of Vanderbilt Mechanical Engineering as an assistant professor in January, 2004. In fall 2010, he got promoted to be an associate professor.

Dr. Li's research interest includes Micro/Nano scale energy and molecular transport phenomena, Micro/Nanofabrication techniques, Molecular dynamics and Monte Carlo Simulation, and Micro/Nanofluidics. He has 46 refereed journal publications in well-respected journals such as *Nature Nanotechnology*, *Nano Letters*, *Physical Review Letters*, *Applied Physics Letters*, *Small*, *Lab-on-a-chip*, *Analytical Chemistry*, and *Journal of Heat Transfer*. His journal publications have been cited for more than 2400 times, with 9 publications each cited over 100 times. He is recipient of National Science Foundation Career Award in 2007.

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Li Qiao

Assistant Professor of Aeronautics and Astronautics

Dr. Li Qiao has a Ph.D. in aerospace engineering from the University of Michigan and M.S. and B.S. degrees in engineering mechanics from China's Tsinghua University.

Dr. Qiao was a graduate research assistant in the Department of Aerospace Engineering at the University of Michigan. Her dissertation involved fundamental understanding of combustion chemistry and dynamics in microgravity. She received several awards at UM, including the prestigious Rackham Predoctoral Fellowship. She was also a two-time recipient of the Amelia Earhart Fellowship by the Zonta International Foundation.

Dr. Qiao's research interests focus on both fundamental understanding and new technology development in the areas of combustion, propulsion, and sustainable energy.

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Nanoscale Energy Transport and Conversion Laboratory

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About the PI:



Professor Ruan received his B.S. and M.S. in Engineering Thermophysics from Tsinghua University in 2000 and 2002, respectively. He received another M.S. in Electrical Engineering in 2006 and Ph.D. in Mechanical Engineering in 2007 from the University of Michigan at Ann Arbor. He then joined Purdue as an assistant professor. Dr. Ruan received the prestigious NSF CAREER Award in 2012. He was an Air Force Summer Faculty Fellow at the Wright Patterson Air Force Base at Dayton, Ohio in 2010 and 2011. He is an Editorial Board Member for *Scientific Reports*, a journal published by the Nature Publishing Group.

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e-mail: ruan@purdue.edu

[see people in the group]

News:

Positions open: We may have positions for postdoc fellows and Ph.D. students. Please check here.

08/2012: **Student Awards**. New students Xiangyu Li and Zexi Lu have received the Ross Fellowship from the School of Mechanical Engineering. Congratulations, Xiangyu and Zexi!

08/2012: **Welcome New Students**. Tianli Feng, Xiangyu Li, Zexi Lu, Jingjing Shi, and Zuyuan Wang have joined our group. Welcome!

BU Mechanical Engineering

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Chuanhua Duan, Ph.D.



Assistant Professor

Ph.D., University of California, Berkeley

phone: (617) 353-3270
email: duan@bu.edu
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website: http://www.bu.edu/neft

Research Interests

Energy conversion and storage * Micro/nanofluidics * Phase change heat transfer

Chuanhua Duan focuses his research on studying enhanced ion/molecule transport using ordered micro/nanostructures, for both fundamental understanding and practical applications in energy conversion/storage and biomolecular sensing/separation.

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Principal Investigator



Renkun Chen, Ph.D.

Assistant Professor, UC San Diego
Department of Mechanical and Aerospace Engineering
Materials Science and Engineering Program
Ph.D. in Mechanical Engineering, UC Berkeley, 2008
B.S. in Engineering Thermo-Physics, Tsinghua University, China, 2004

Prof. Renkun Chen joined the UCSD faculty in November 2009. Chen received his Ph.D. in Mechanical Engineering at the UC, Berkeley, where his research in Prof. Arun Majumdar's group in focused on nanoscale thermal energy conversion. Subsequently, Chen continued working as a post-doctoral fellow in Prof. Majumdar's group, where he investigated nanoscale phonon transport physics using advanced nanofabrication tools.